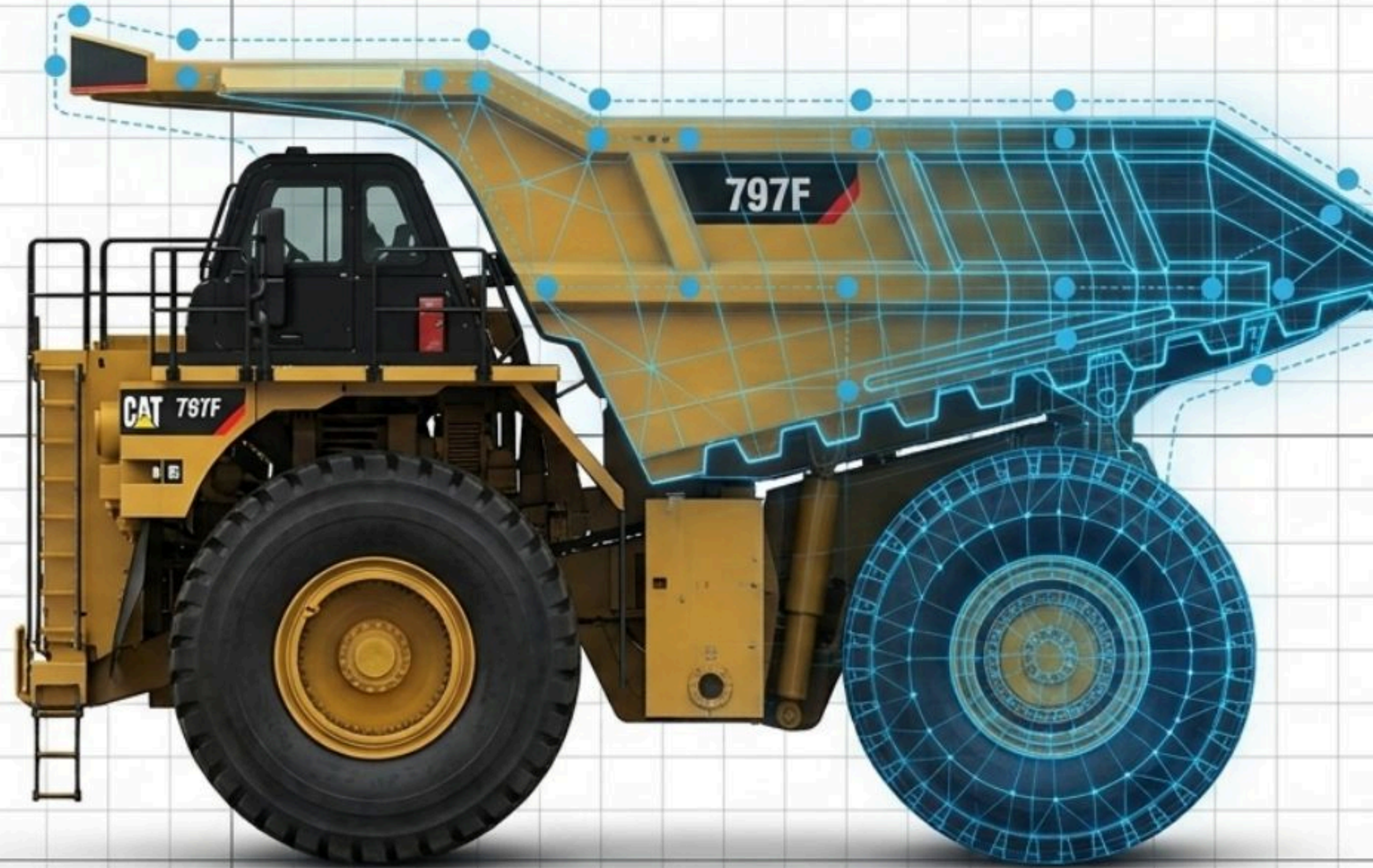


AVOID CARRY-BACK FROM DUMP TRUCKS IN MINING OPERATIONS

Introducing PHANOVEX:
Predictive Carry-Back
Intelligence System



THE CARRY-BACK CHALLENGE: PROBLEM, IMPACT, & GAPS

Carry-back occurs when material remains in the truck bed after dumping, reducing payload capacity, increasing equipment wear, and causing operational inefficiencies. Current dump processes often lead to incomplete material discharge. This issue impacts productivity, safety, and maintenance costs in autonomous and manual mining operations.

PROBLEM DESCRIPTION



Material remaining in truck beds after dumping reduces effective payload capacity



Leads to repeated haul cycles and operational delays



Causes uneven load conditions affecting vehicle performance



Requires manual intervention for cleaning

OPERATIONAL IMPACT



Productivity loss
→ Reduced material transport efficiency



Higher fuel consumption
→ Extra trips required



Maintenance increase
→ Tray and hoist wear



Safety risks
→ Material falling during transit

CURRENT OPERATIONAL GAPS



No reliable real-time detection of carry-back during dump cycles



No prediction of carry-back risk before dumping occurs



Harsh mining conditions reduce effectiveness of existing monitoring methods



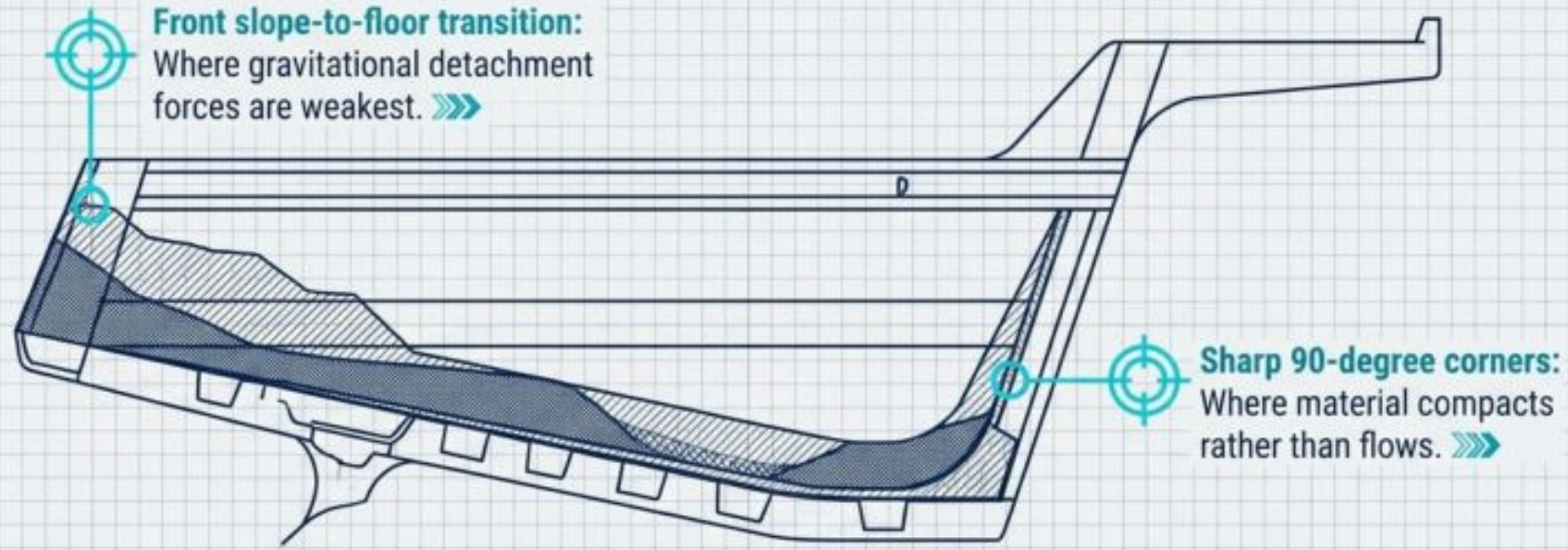
Maintenance remains reactive rather than predictive



Lack of intelligent, data-driven dump optimization

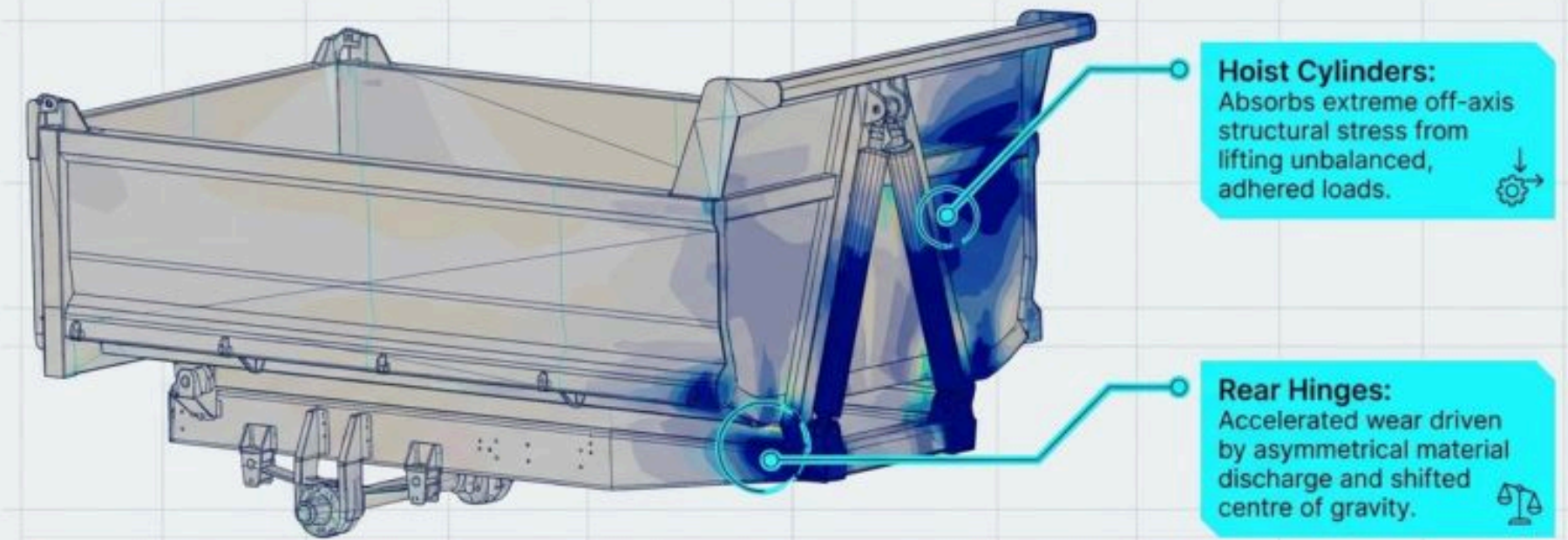


Where and How Residual Material Accumulates



Carry-back is a progressive accumulation, not a static event. The thin residual layer from cycle one alters the surface wettability and roughness, drastically increasing the probability of severe adhesion in cycle two.

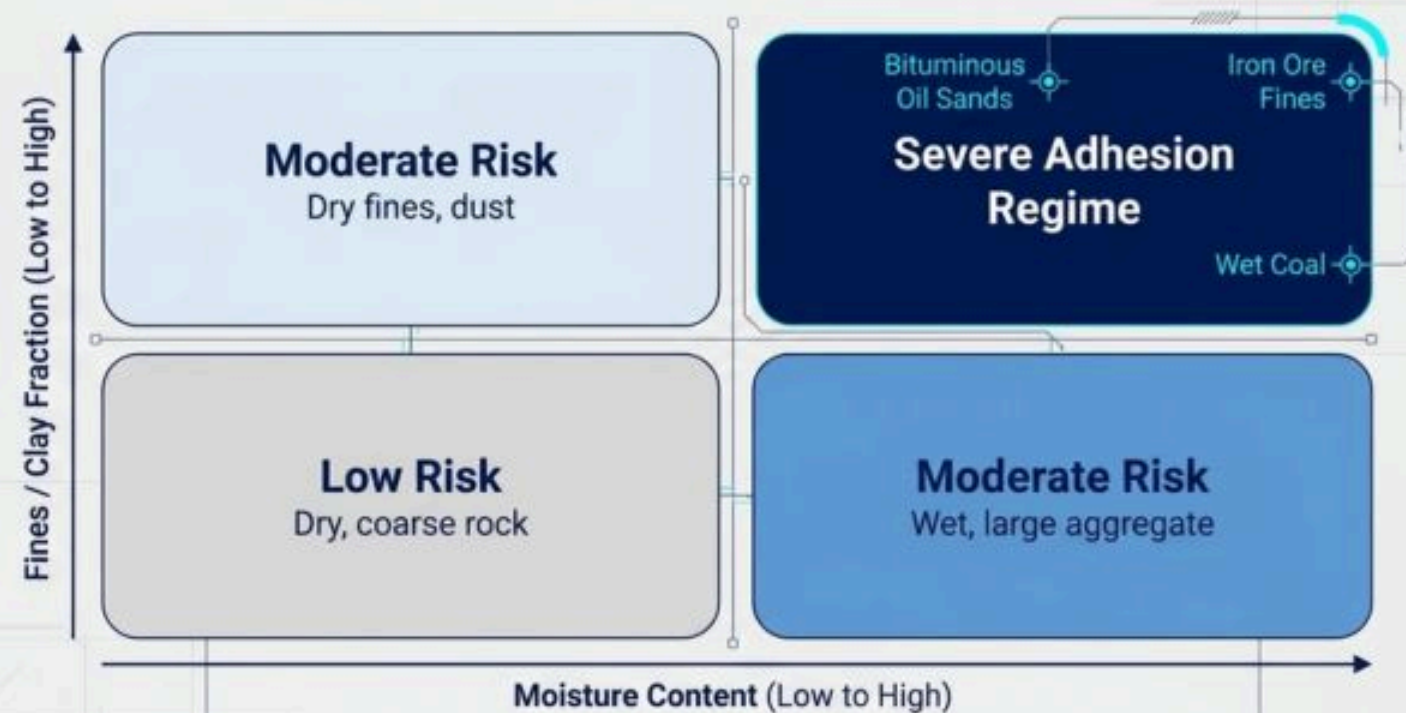
Asymmetrical Stress Accelerates Mechanical Degradation



Temporarily shifts vehicle centre of gravity.

Blueprint of Inefficiency

Cohesive Ores Fundamentally Alter the Sticking Profile

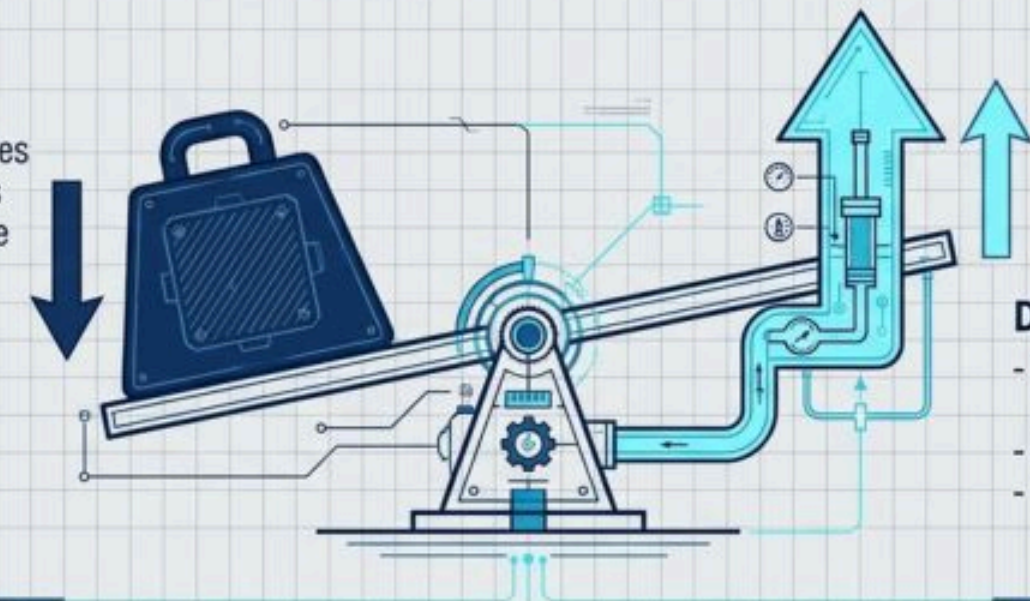


Impact and sliding dynamics fail when moisture and fines peak, creating severe hang-up on impact plates and truck bodies.

The Dynamic Equilibrium Between Adhesion and Detachment

Adhesion Forces

- Capillary liquid bridges
- van der Waals forces
- Compaction pressure
- Ice-like bonding



Failure Occurs in the Adhesion-Dominated Regime.

When cohesive forces overcome gravitational detachment during the tipping flow, the cycle inevitably results in material hang-up.

PHANOVEX: Predictive Carry-Back Intelligence System

See it. Predict it. Prevent it.

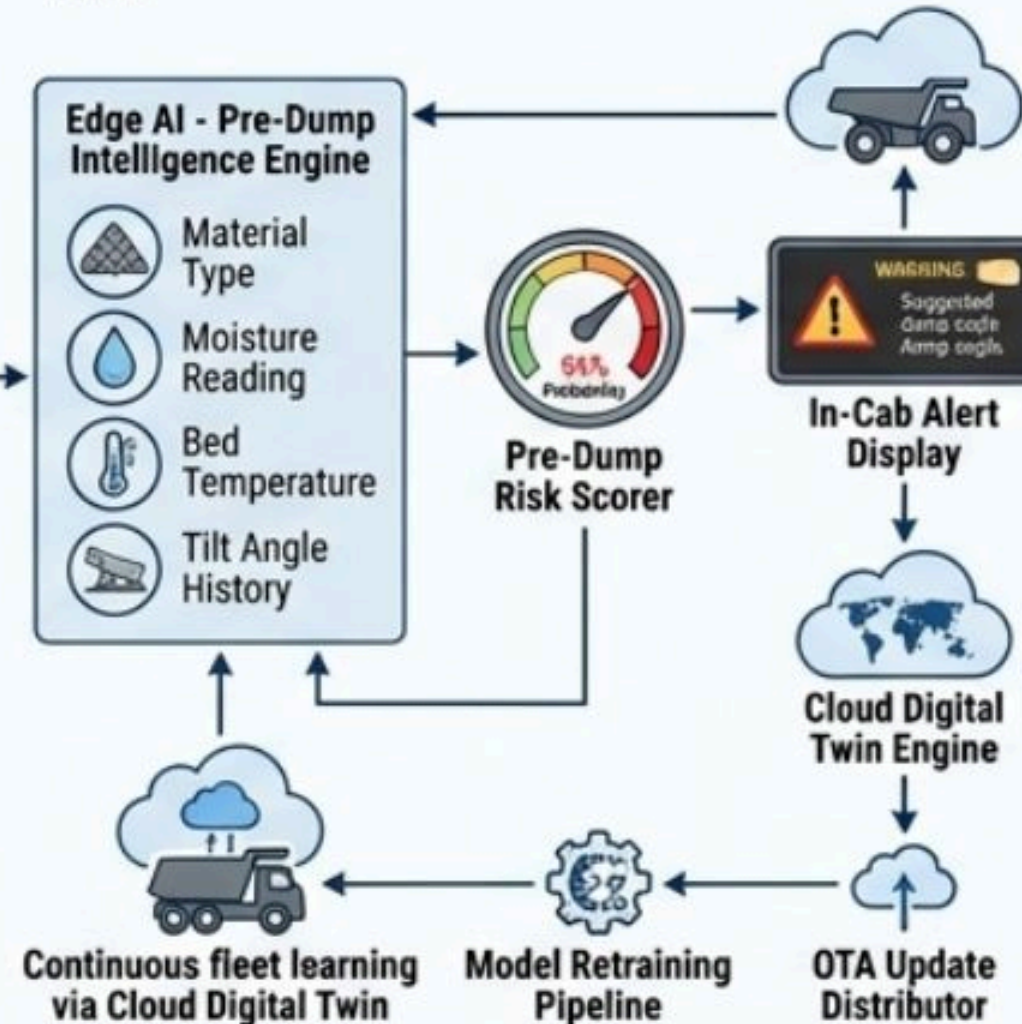
Intelligent Detection

- Multi-sensor fusion platform
- IP68 Stereo Vision
- Load Cell deltas
- Hydraulic pressure
- Moisture index sensor



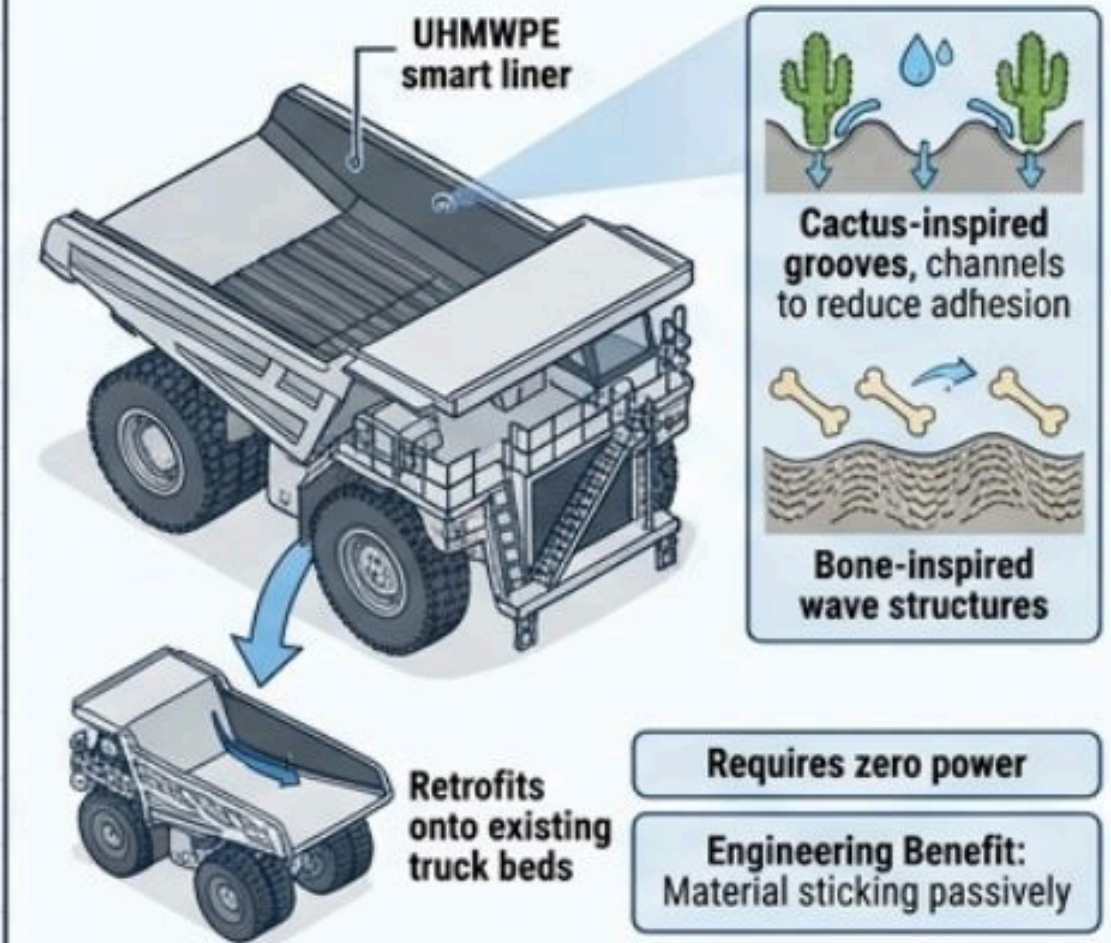
Smart Feedback & Learning

- Edge AI generates pre-dump risk score
- In-cab operational alerts
- Continuous fleet learning via Cloud Digital Twin

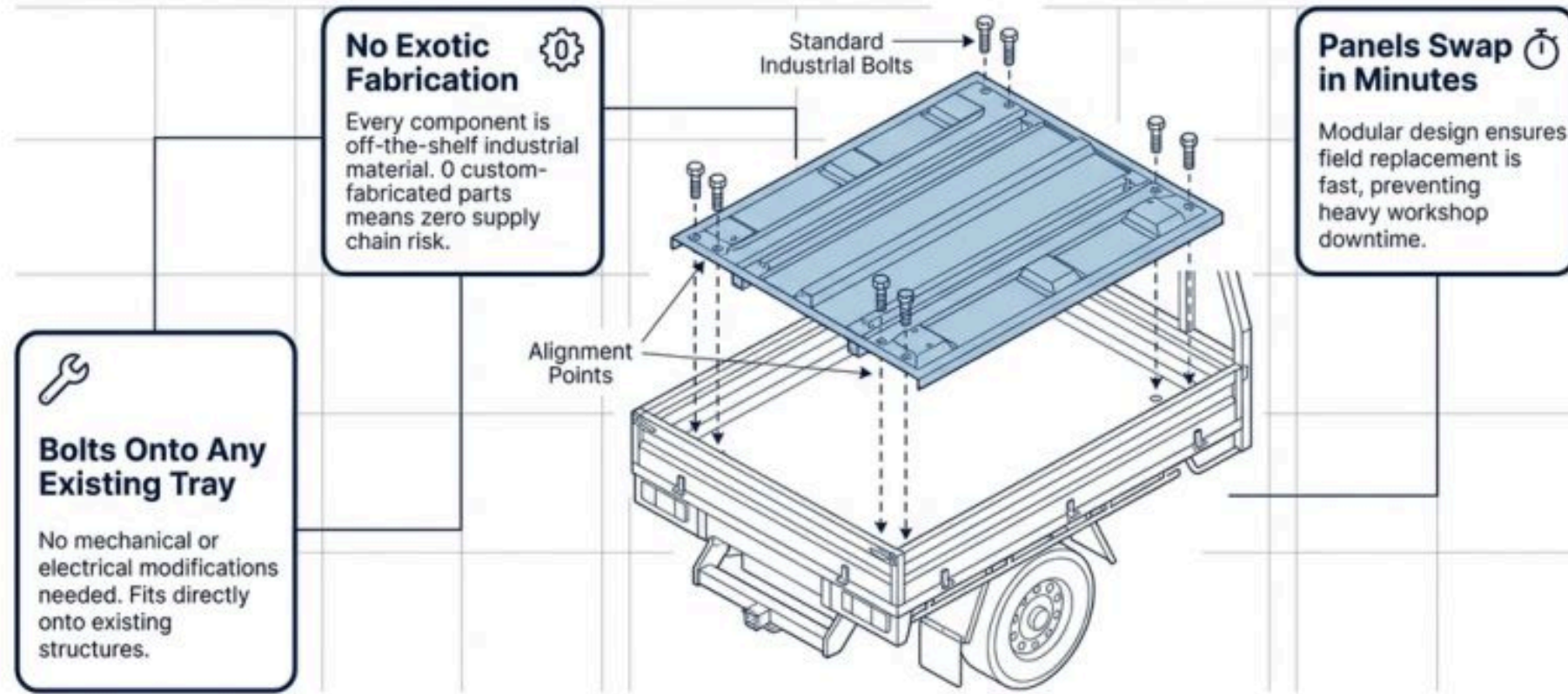


Biomimetic Prevention

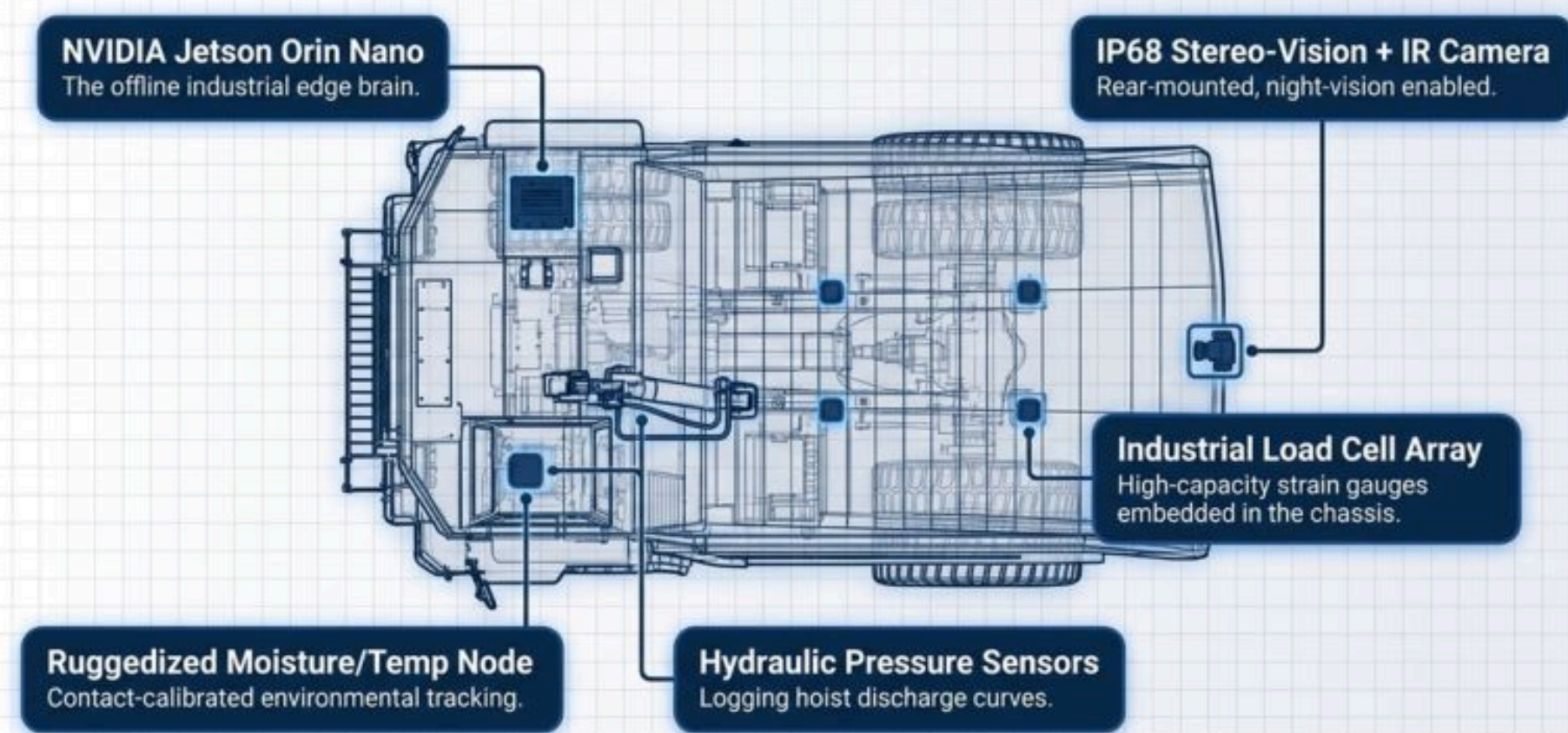
- Nature-inspired surface engineering (UHMWPE smart liner)
- Requires zero power
- Retrofits onto existing truck beds



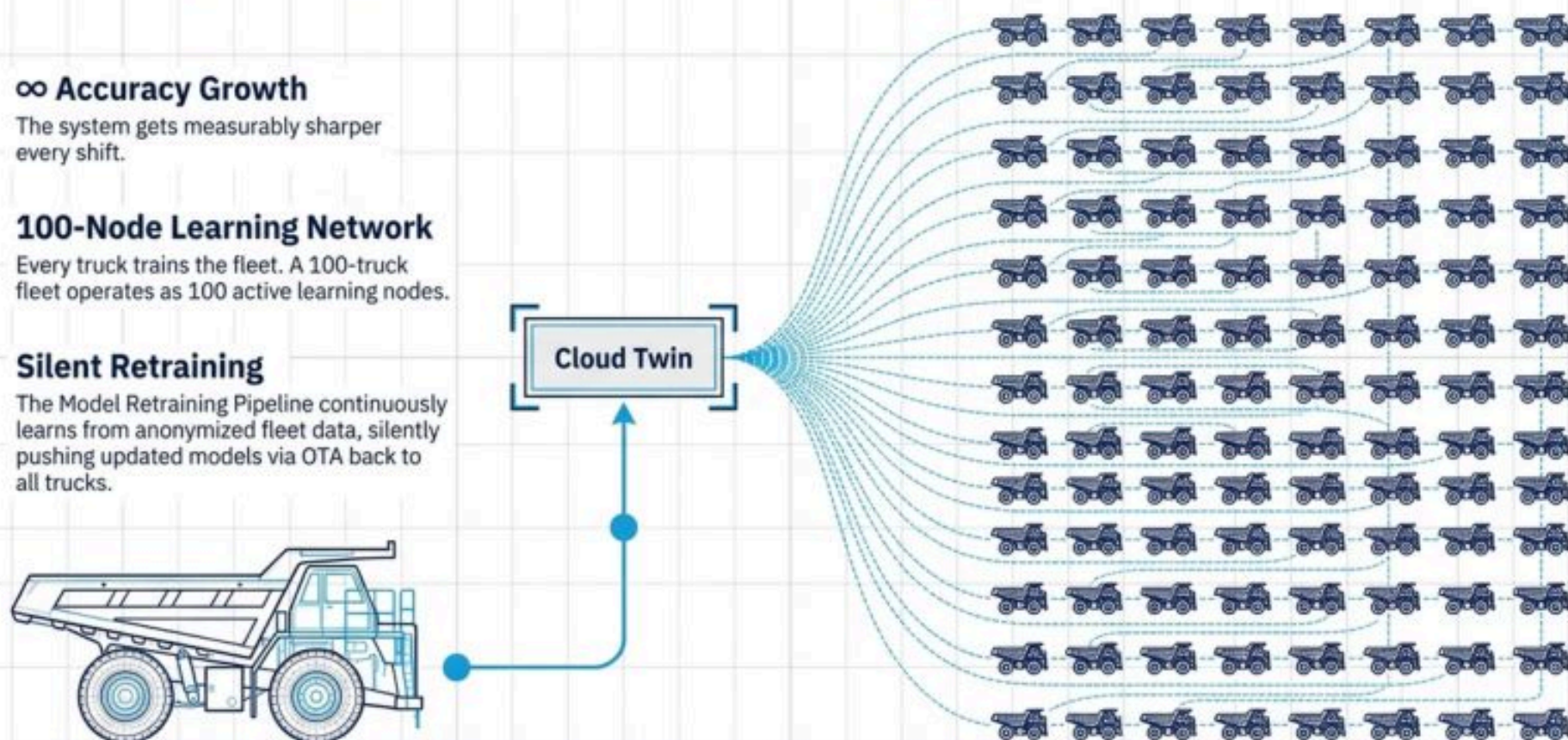
Deployment Economics: Built for Retrofit



Sensor Fusion Layer



The Compound Effect: Fleet Scale Learning



Edge AI & The Pre-Dump Risk Score



Fires Before The Dump

Generates a probability severity score in the seconds before the bed moves, not after.

Fully Offline Compute

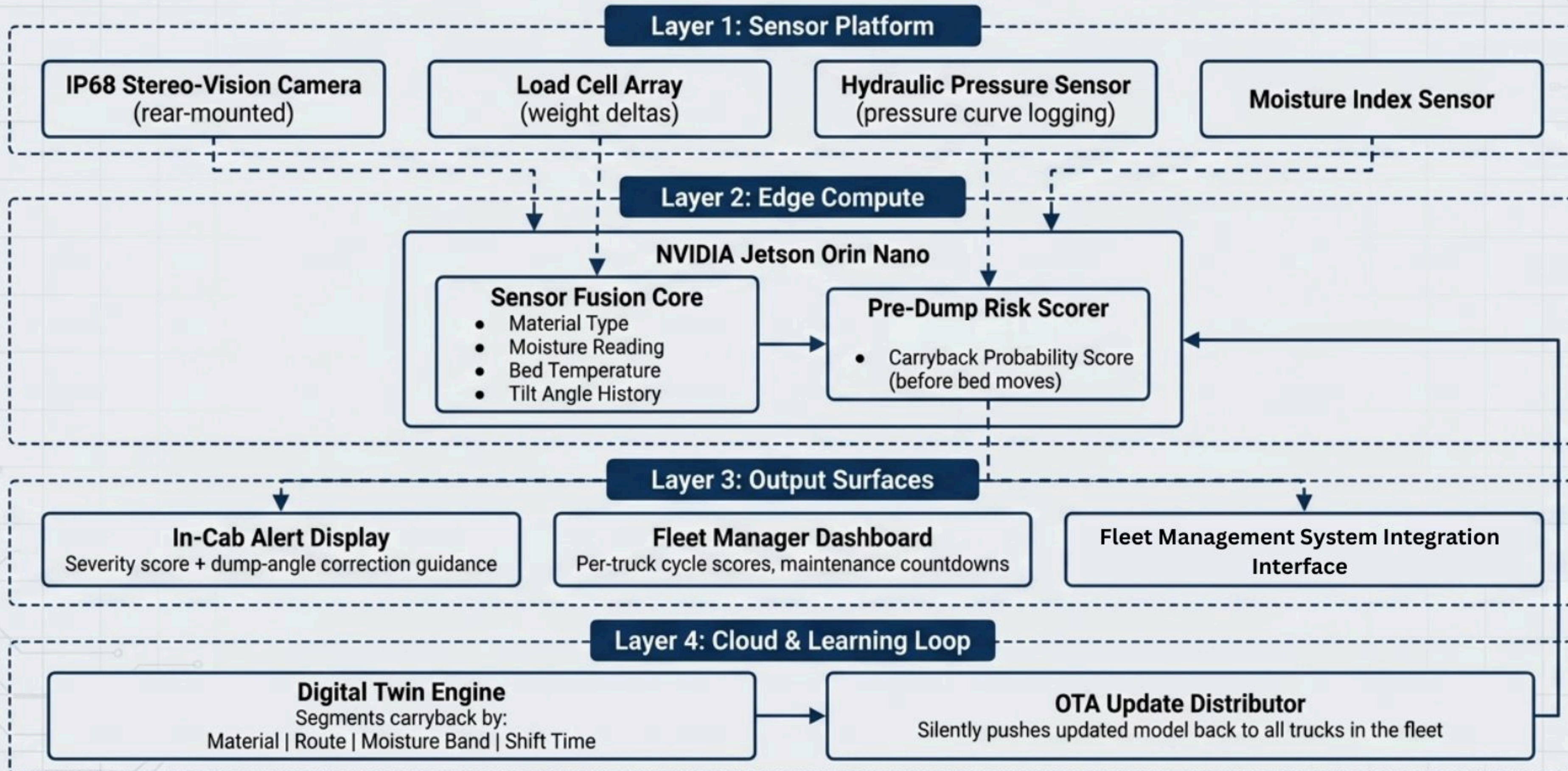
Runs purely on the sensor fusion core for zero-latency performance in disconnected mining environments.

Tri-Surface Output

Delivers simultaneous alerts to the In-Cab Display, Fleet Manager Dashboard.

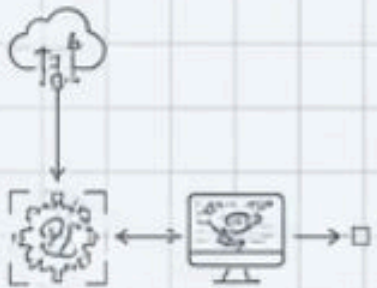
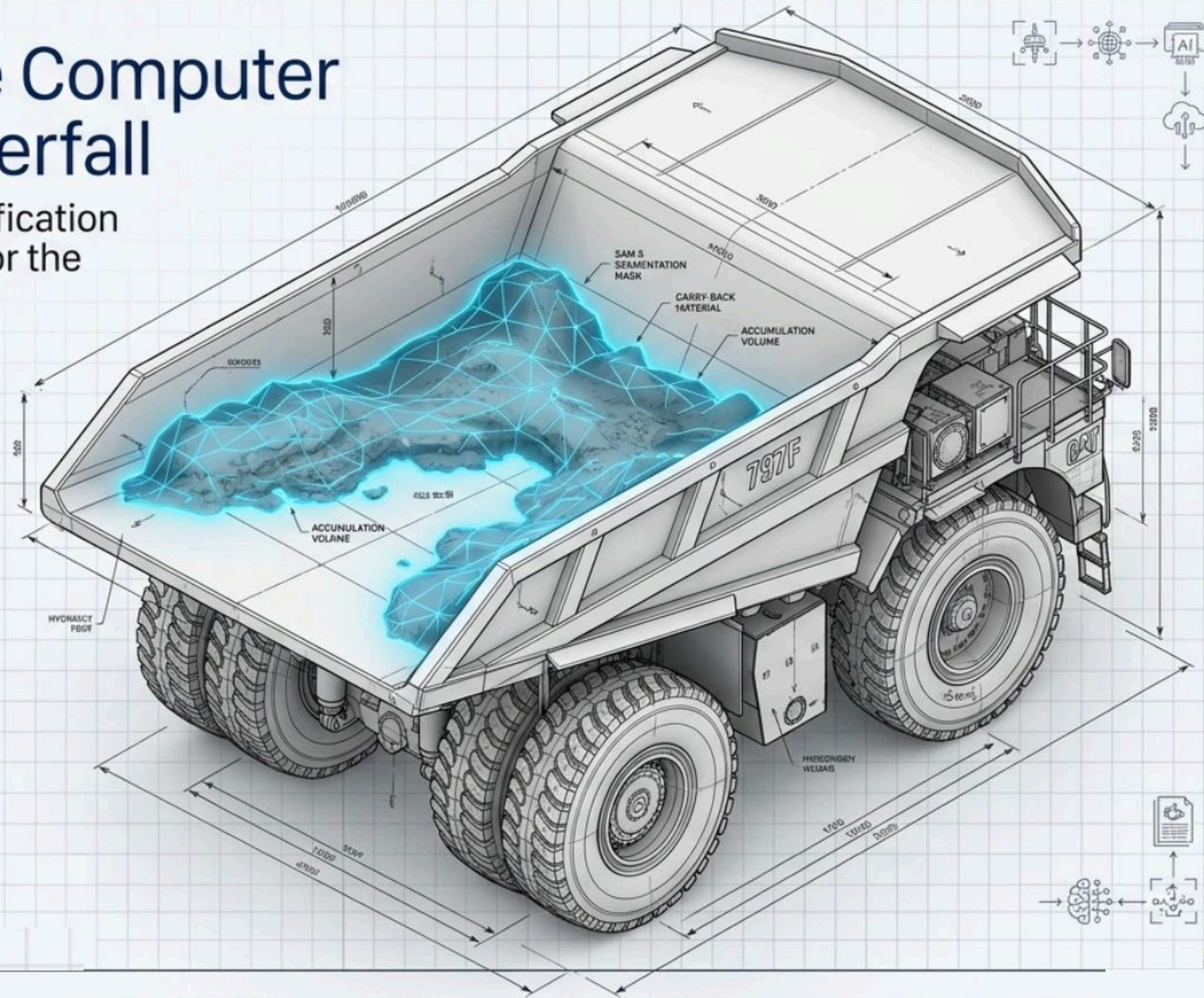
The Phanovex Master Blueprint

See it. Predict it. Prevent it.



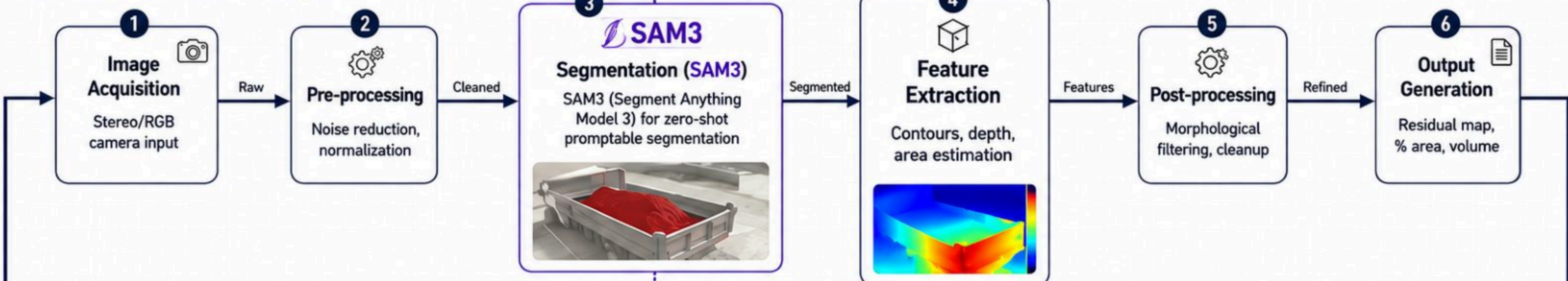
Engineering the Computer Vision Data Waterfall

Pixel-Level Carry-Back Quantification and Edge Inference Pipeline for the Phanovex Architecture.



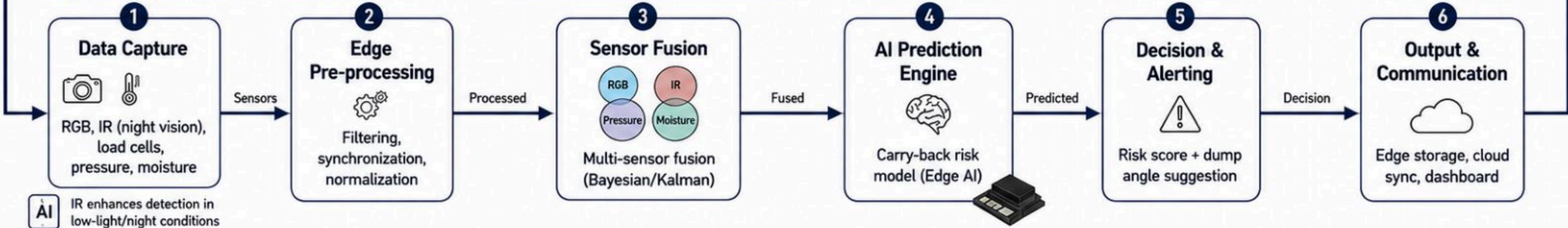
System Architecture: The Data Waterfall (Updated with SAM3)

Image Processing Pipeline



- Why SAM3?**
- State-of-the-art zero-shot segmentation
 - Robust across lighting, clutter & domains
 - Promptable for targeted segmentation

Edge Computing Workflow



AI IR enhances detection in low-light/night conditions

Hardware
Edge device (Jetson),
ARM, Linux

Models
CNN, YOLO,
SAM3, Sensor Fusion

Performance
Latency < 5s, Accuracy > 90%,
Low false alarm rate

Camera = Acquisition

Gear = Processing

Brain = AI

Cloud = Cloud

Chip = Edge

Sensors = Inputs

SAM3 Segmentation model updated to SAM3 (Segment Anything Model 3) for improved generalization, accuracy, and robustness in real-world mining environments.

Optical Normalisation in Harsh Environments



Contrast Multiplier: 1.4

Enhances local texture differentiation between dust-covered metal and actual dirt.

Clip Limit: 2

Prevents overexposure on reflective metallic surfaces.

Normalise Brightness: True

Corrects for severe shadowing and variable sun angles inside the deep truck tray.

These parameters actively separate material texture from the dusty tray background before inference

The Algorithmic Shift: Segment Anything Model 3

Traditional Object Detection



- **Model Architecture:** Conventional CNN-based detector.
- **Mechanism:** Region proposal and bounding box regression.
- **Output Format:** Bounding Box Coordinates (Xmin, Ymin, Xmax, Ymax).

SAM 3 Zero-Shot Segmentation



- **Model Architecture:** Block roboflow_core/sam3@v3 via sam3/sam3_final.
- **Mechanism:** Promptable concept segmentation using text inputs ('dirt', 'mud', 'ore').
- **Output Format:** Run-Length Encoded (RLE) Masks.

The Advantage: RLE masks isolate exact material pixels, providing the mathematically rigorous foundation required for accurate volume proxy calculations.

Calibrating the Segmentation Engine



Confidence Threshold

Why 0.04 Confidence?

Intentionally hyper-sensitive. In mining, missing a thin, adhesive clay film is a systemic failure. We accept a higher false-positive rate to guarantee detection of faint, low-contrast carry-back.



NMS IoU Threshold

Why 0.98 NMS IoU?

Non-Maximum Suppression is nearly disabled. By only suppressing nearly identical masks, we preserve highly fragmented, overlapping material remnants critical for true area calculation.

Post-Processing Mathematics: Area Percentage

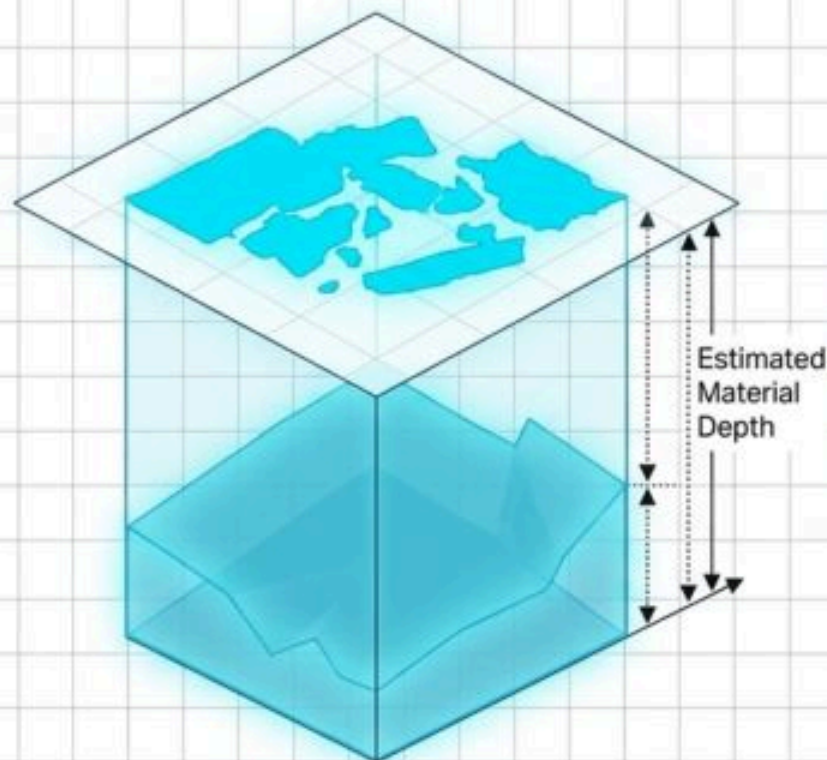
$$\frac{\text{[Sum of Material Mask Pixels]}}{\text{[Tray Region Pixels]}} = \text{Residual Area \%}$$
A diagram illustrating the formula for Residual Area Percentage. It shows a blue irregular shape representing the 'Sum of Material Mask Pixels' above a horizontal line, and a blue trapezoid representing the 'Tray Region Pixels' below the line. An equals sign follows, leading to 'Residual Area %'.

SAM 3 delivers the RLE mask, but the pipeline must restrict analysis to the operational zone. The formula requires automatic tray region isolation and mask clipping to generate a mathematically sound severity proxy.

Translating 2D Area to a Volume Proxy

A single 2D image cannot directly measure volume without stereoscopic depth assumptions.

Our Pipeline Solution: We utilize the segmented area percentage as our primary severity proxy, multiplying Area $\times$ Estimated Material Depth by class.



Future implementation paths include stereo/depth camera integration to measure actual residual height.

Establishing Rigorous Validation Metrics

mAP@50-95

The critical benchmark for segmentation boundary accuracy across varying IoU thresholds.

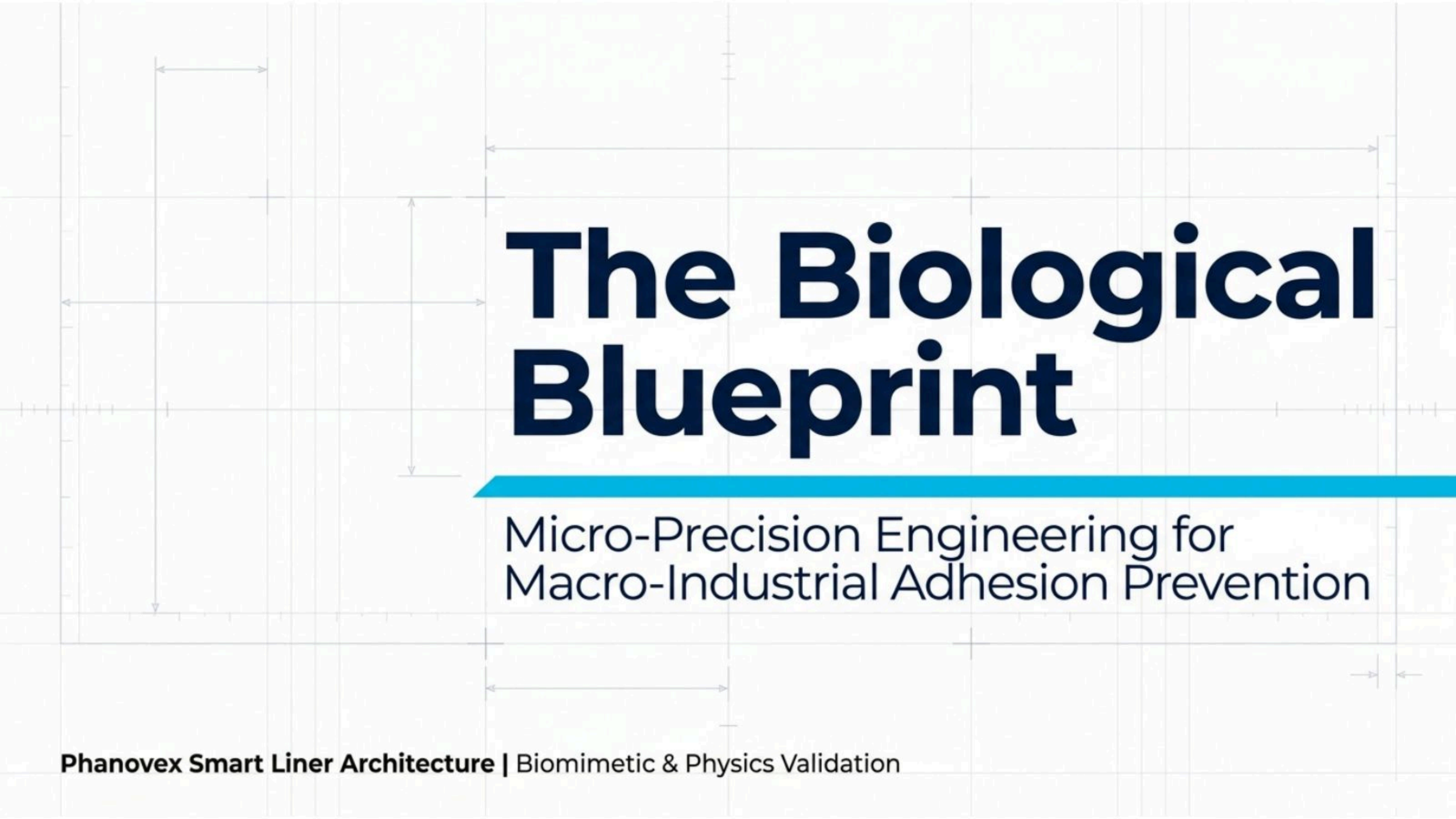
Precision vs. Recall

Balancing false alarms (shadows/dust) against the operational danger of missed carry-back (recall prioritized).

Dice Coefficient

Essential for evaluating irregular, non-convex material patches where standard pixel accuracy is misleading.

Note: Current zero-shot prototyping will transition to these measured benchmarks against a human-annotated validation set.

The background of the slide is a technical drawing grid. It features a light gray grid with horizontal and vertical lines. Overlaid on this grid are several dimension lines, represented by thin gray lines with arrows at their ends, indicating measurements in various directions. The main title is positioned on the right side of the slide, with a thick blue horizontal bar located directly beneath it.

The Biological Blueprint

Micro-Precision Engineering for
Macro-Industrial Adhesion Prevention

Granular Physics Validation: DEM & LIGGGHTS

The Theory

LIGGGHTS is an advanced open-source Discrete Element Method (DEM) simulation tool.

We utilized it to model the dynamic adhesion, mechanical stress, and flow of sticky ores (e.g., iron ore fines).

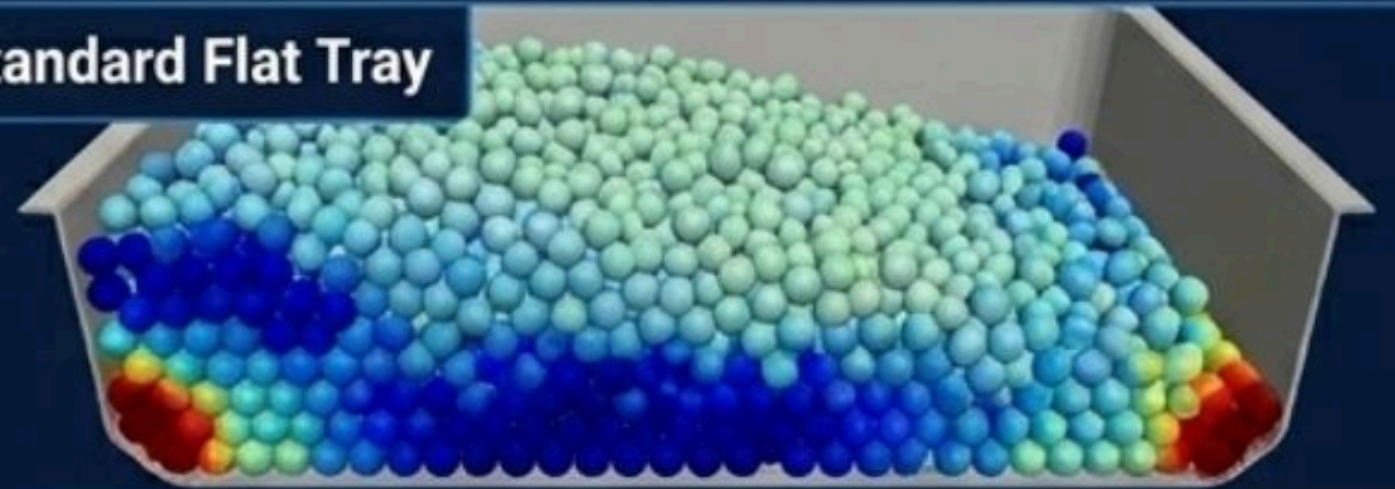
The Mathematical Engine:

Newton's equations of motion for granular cohesive flow

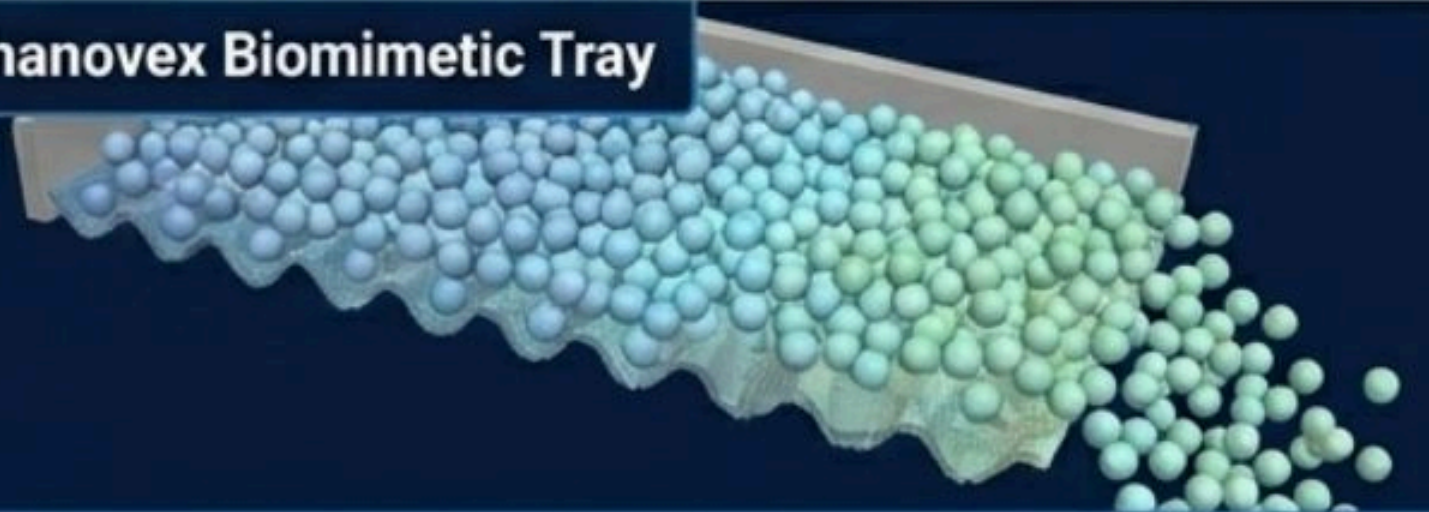
$$m_i \frac{d^2 x_i}{dt^2} = F_i^{\text{contact}} + F_i^{\text{gravity}} + F_i^{\text{cohesion}}$$

The Simulated Output

Standard Flat Tray



Phanovex Biomimetic Tray



Engineering Benefit: Allowed virtual testing of thousands of dump cycles, proving our specific liner geometry successfully distributes compaction stress and prevents deep particle embedding.

Flow Dynamics Validation: CFD & OpenFOAM

The Theory

OpenFOAM is an advanced C++ Computational Fluid Dynamics (CFD) toolbox.

We deployed it to simulate the macroscopic fluid-like discharge flow of highly viscous, moisture-heavy materials over our custom liner grooves.

The Mathematical Engine:
Navier-Stokes equations modeling
multiphase flow dynamics

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

The Simulated Output



Engineering Benefit: Verified that our micro-grooves effectively disrupt the continuous contact area, breaking capillary bridges and van der Waals adhesion forces during the tipping phase.

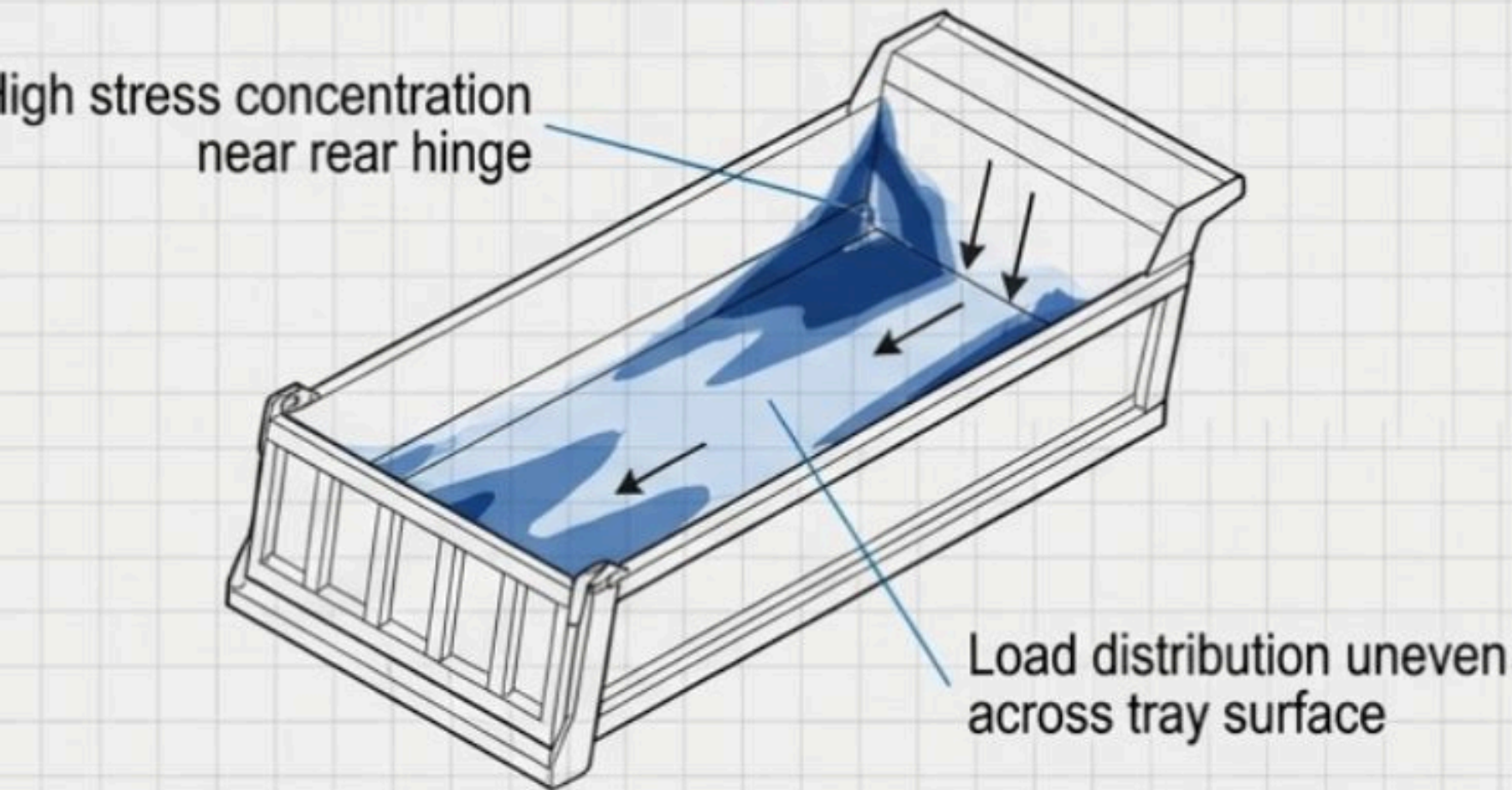
Evolutionary R&D: Achieving Release Without Energy

Flora and fauna survive extreme environments not through active energy consumption, but through passive geometric precision. By mapping these natural blueprints onto UHMWPE liners,

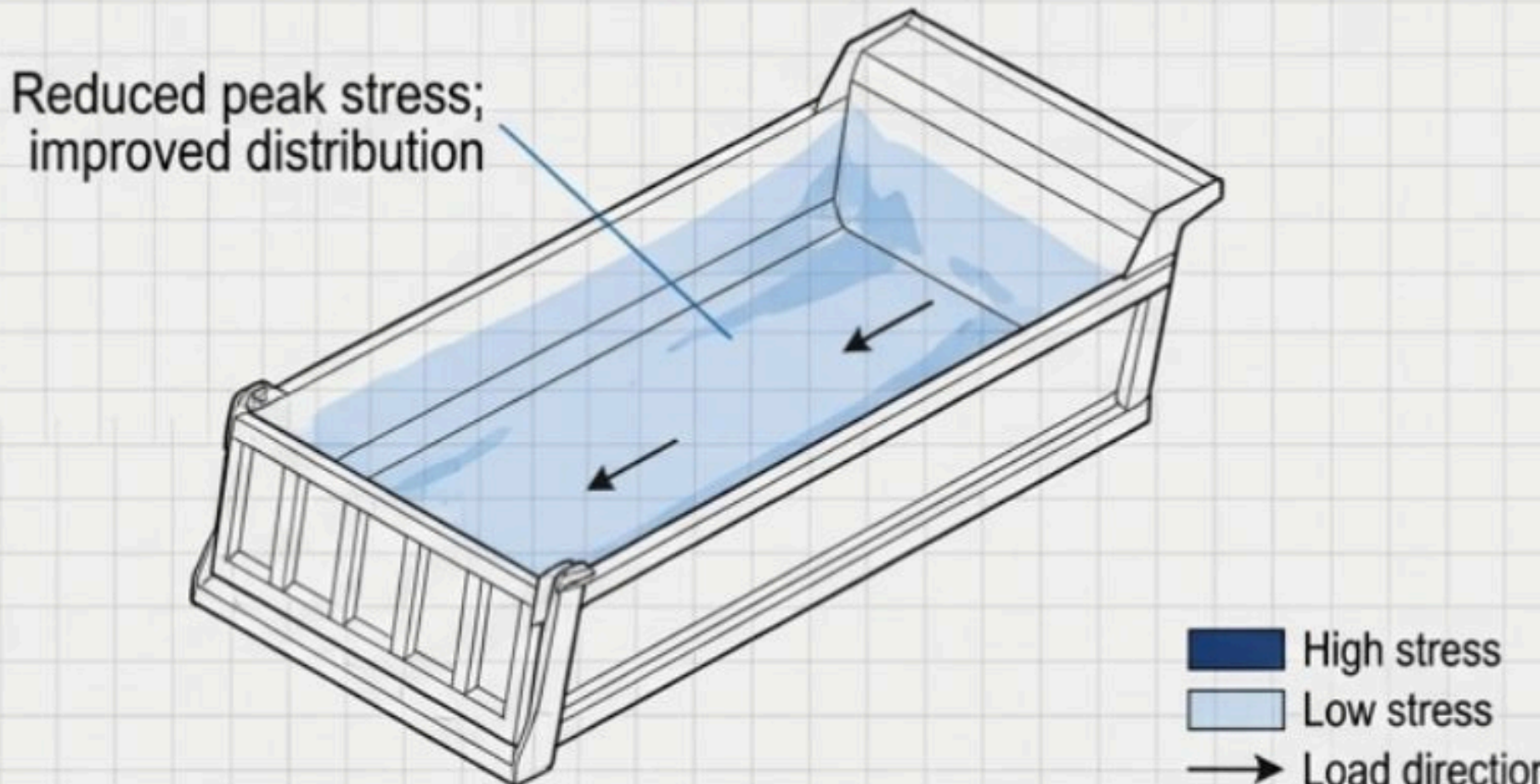
Macro-scale efficiency is solved through meso-scale geometry.

Structural & Material Stress Validation

Before Optimization



After Optimization



Experimental Validation

- Load testing cycles ($N > 10^4$)
- Abrasion resistance evaluation
- Repeated dump cycle analysis

Computational Analysis

- Finite Element Analysis (FEA) on tray structure
- Stress distribution mapping under load conditions
- Material response under cyclic loading

Performance Metrics

- Stress reduction: ↓ peak stress regions
 - Wear rate: ↓ surface degradation
 - Structural load distribution: improved uniformity
- ± error margins considered; Environmental variability accounted (moisture, load type)

Experimental and computational analysis indicate reduced stress concentration and improved material durability under operational conditions.

Field Analysis & Observed Performance

Before Deployment

Localized
adhesion in
rear tray
zones due to
material-
surface
interaction.



- High residual material retention in truck bed
- Non-uniform discharge behavior
- Frequent manual cleaning cycles

Cycle Efficiency ↑

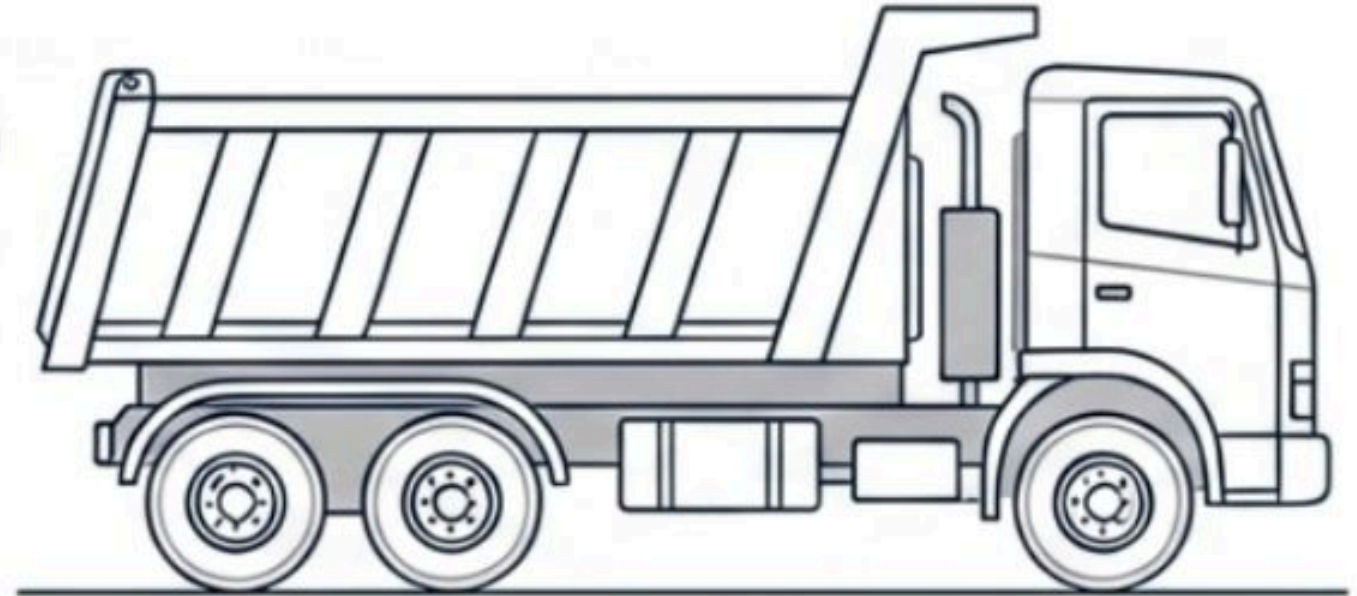
- Reduced number of dump cycles per load
- Improved operational throughput

Maintenance Frequency ↓

- Lower manual cleaning requirements
- Reduced mechanical wear on hoist/tray

After Deployment

Improved
surface
interaction
enables
uniform
material
discharge.

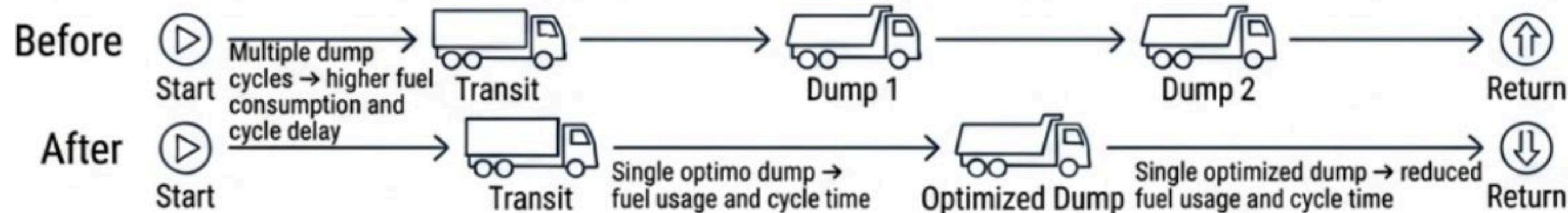


- Reduced residual material accumulation
- Uniform material discharge profile
- Minimal manual intervention required

Payload Utilization ↑

- Higher effective payload per cycle
- Reduced material carry-back losses

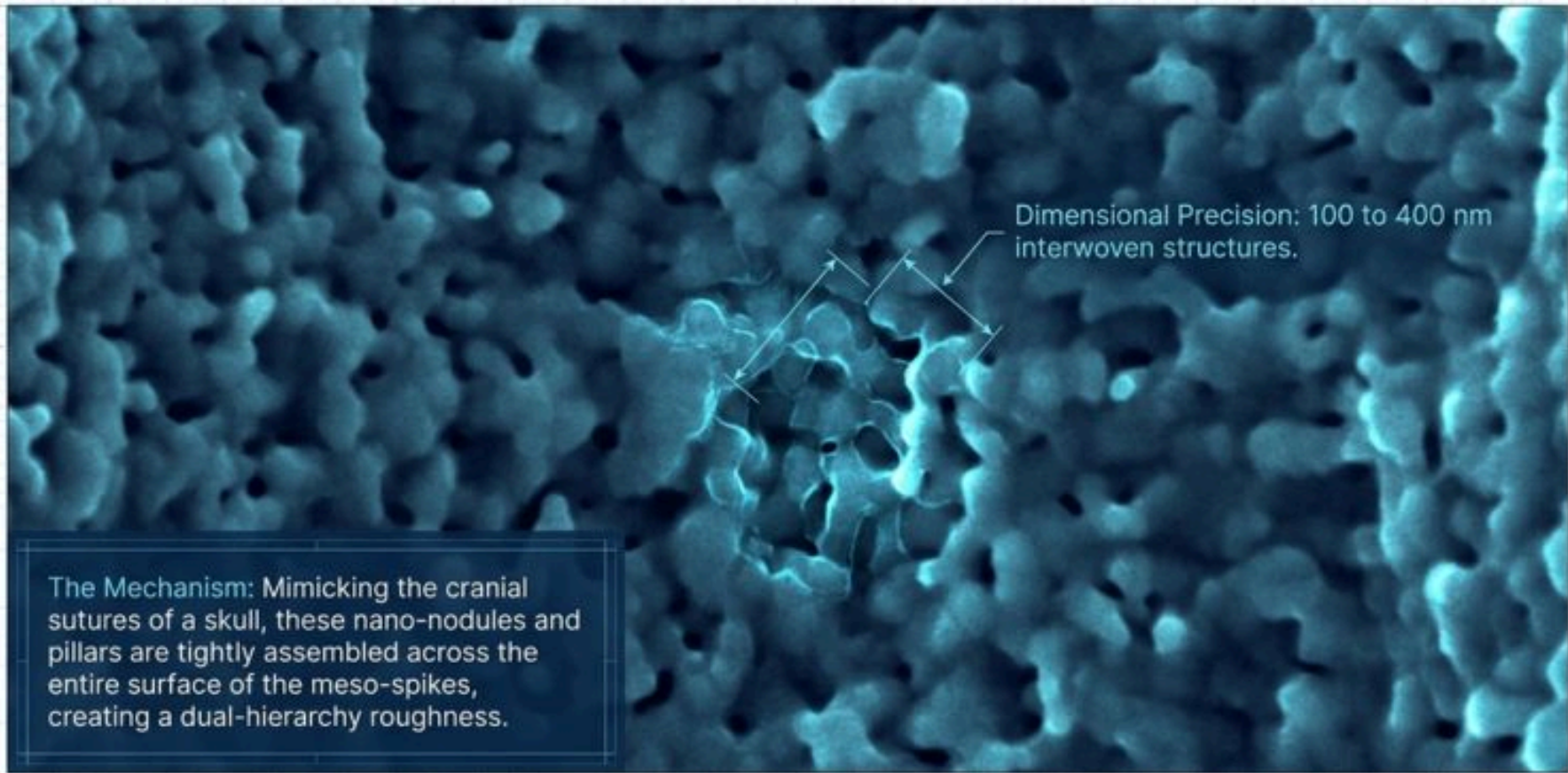
Operational Scenario Comparison



Analysis Insights

- Material adhesion influenced by moisture and surface interaction
- Repeated dumping increases structural stress accumulation
- Improved surface-composite interaction enhances discharge consistency
- Surface-composite interaction influences adhesion and discharge efficiency

Nano-Scale: Bone-Inspired Trabeculae



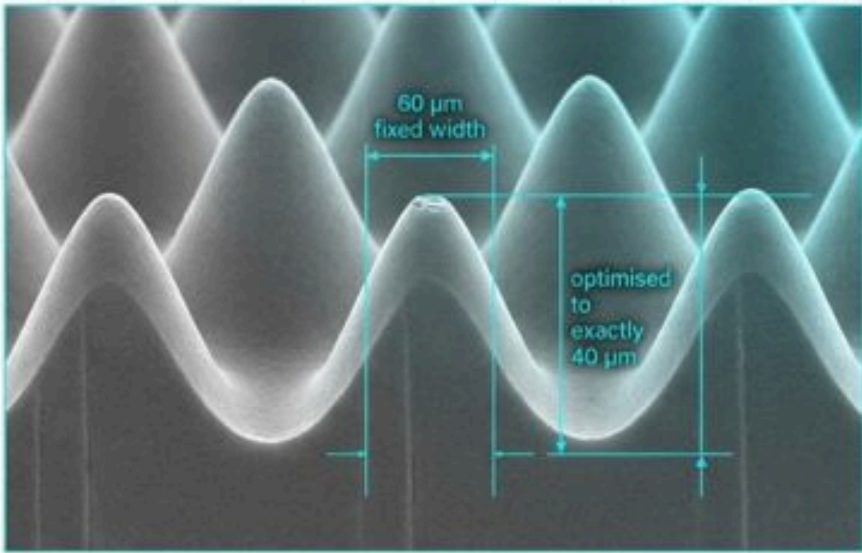
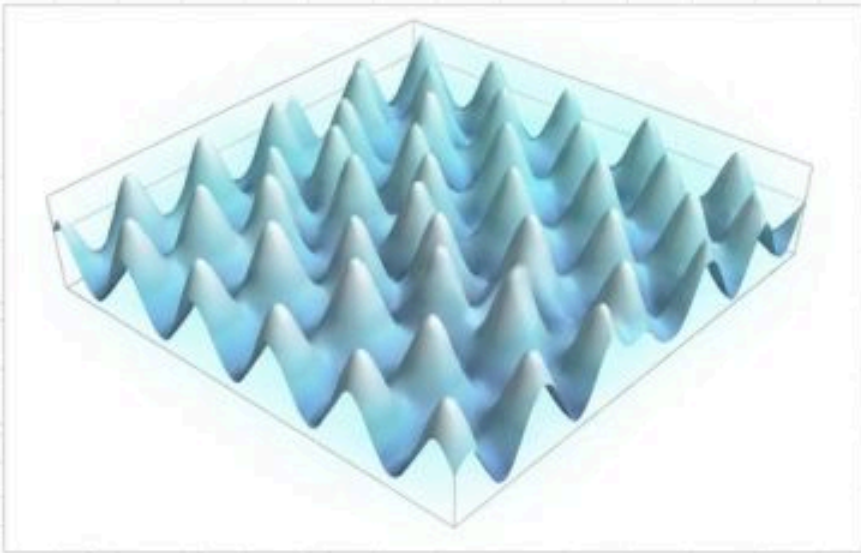
Absorbing Compaction Force



Physical Action
The interlocking nano-pattern catches vertical payload impact and distributes the mechanical stress laterally.

Scientific Outcome
Prevents crack propagation and eliminates localised high-pressure points where granular adhesion is historically strongest.

Meso-Scale: Cactus-Inspired Spikes & Grooves



Dimensional Precision: 60 µm fixed width, optimised to exactly 40 µm in height.

The Mechanism: Creates an oscillating configuration with a repetitive peak-and-valley pattern that physically prevents continuous planar contact between sticky material and the liner.

The Clash of Paradigms: Active vs. Passive

Metric Name	VR Steel Heated Bodies	Phanovex Smart Liner
Core Mechanism	Active 100°C exhaust heat	Passive biomimetic geometry
Energy / Operating Cost	Continuous exhaust drain	Zero ongoing energy
Retrofit Capability	Invasive body redesign & ducting	Seamless bolt-on replacement
Intelligence	Reactive (after sticking occurs)	Proactive & Predictive

Competitive Matrix: Why Existing Solutions Fail

Solution Type	Pre-Dump Prediction	Passive / Zero-Energy?	Fleet Learning?	Retrofit Ease
Manual Wash / Water Cannons	No	No	No	High labor & downtime
VR Steel (Heated Bodies)	No	No (~100°C exhaust gas)	No	Extremely difficult (requires structural redesign)
Suspended Liners (Duratray)	No	Yes (Partial flex)	No	Variable performance on sticky clay
Load Volume Scanners	No (Post-facto)	N/A	Limited	Detects loss, but does not prevent it
Phanovex System	Yes	Yes (Biomimetic)	Yes (100-node OTA)	Standard Industrial Bolts (Minutes)

Current methods react to the symptom.
Phanovex predicts and prevents the root cause passively.

The Lab vs. Pit Hardware Matrix

Variable Tracked	Prototype Hardware	Production Hardware	Upgrade Rationale
Vision & Masking	Cam + USB	IP68 Stereo + IR	Dust penetration, zero-light ops.
Compute & AI	Laptop Host	Jetson Orin Nano	5-15W low-power edge inference, zero-latency offline.
Weight & Force	4x FSRs	Industrial Load Cells + Hydraulics	Surviving 450+ kPa operational loads.
Environmental	2x DHT22	Rugged Moisture Index/Temp	Direct material contact in freezing (-20°C) conditions.

Layer 3: Compounding Accuracy Across the Fleet Network

From Node to Network:

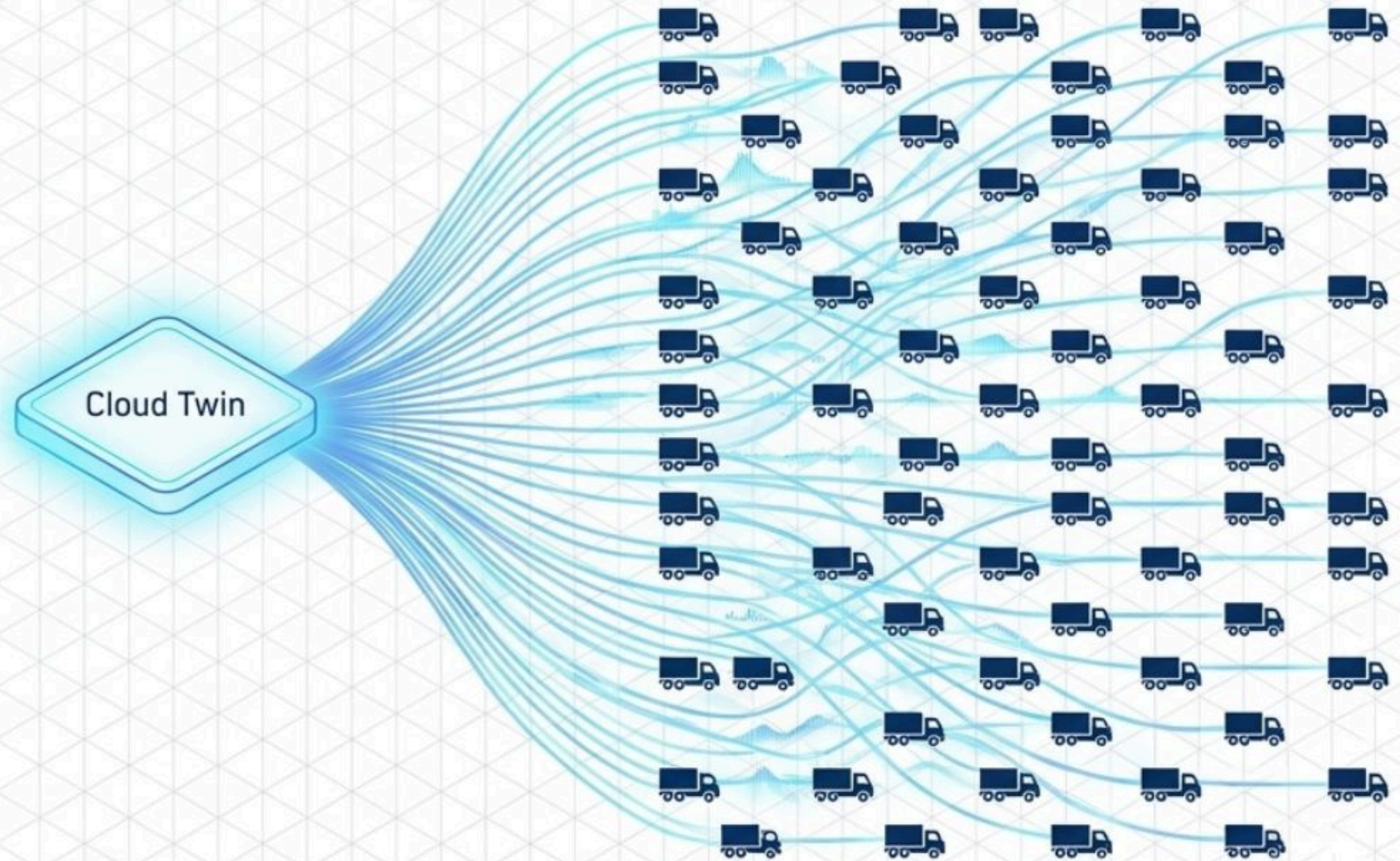
A 100-truck fleet operates as 100 active learning nodes.

Asynchronous Data Sync:

When trucks drive out of deep-pit dead zones and restore Wi-Fi/LTE connectivity, anonymized load and dump data automatically syncs to the central Cloud Digital Twin.

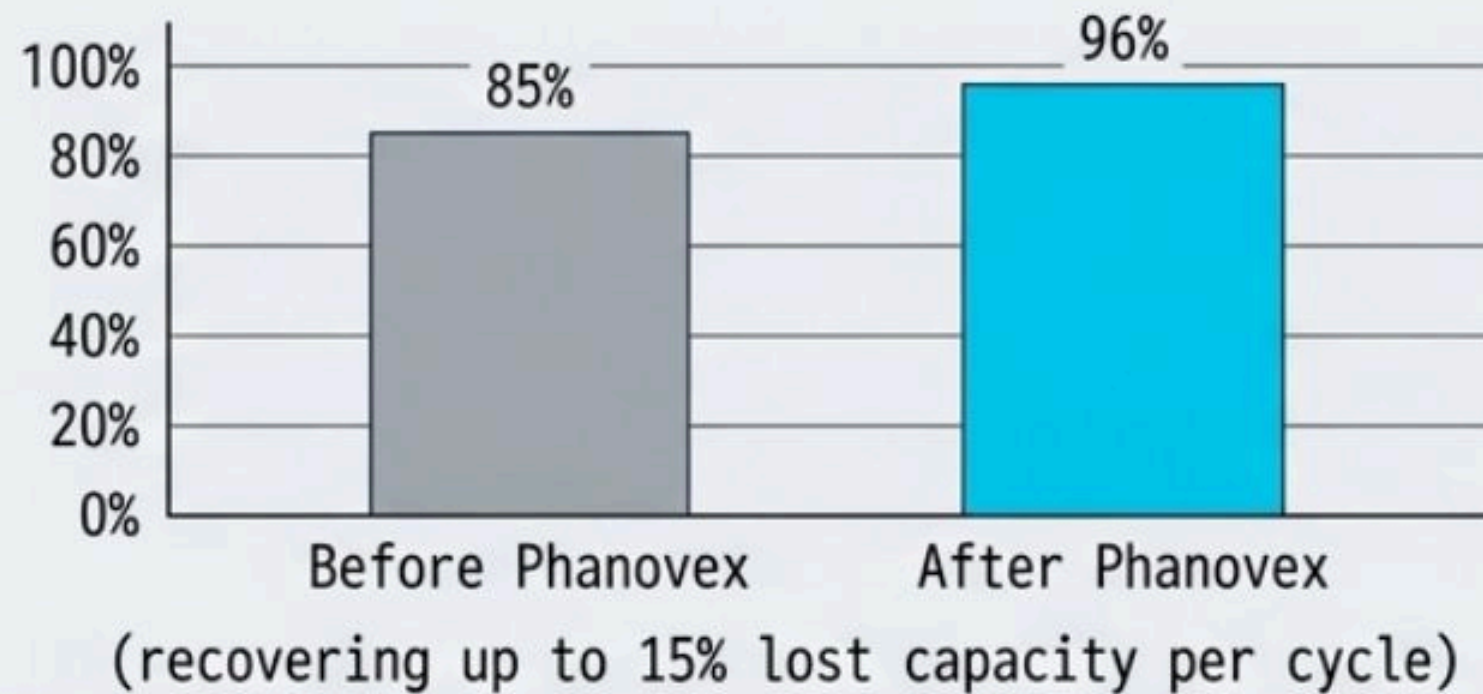
Global Pattern Recognition:

The Cloud model segments carry-back incidents by material type, haul route, weather conditions, and shift time to uncover hidden macroscopic operational patterns.

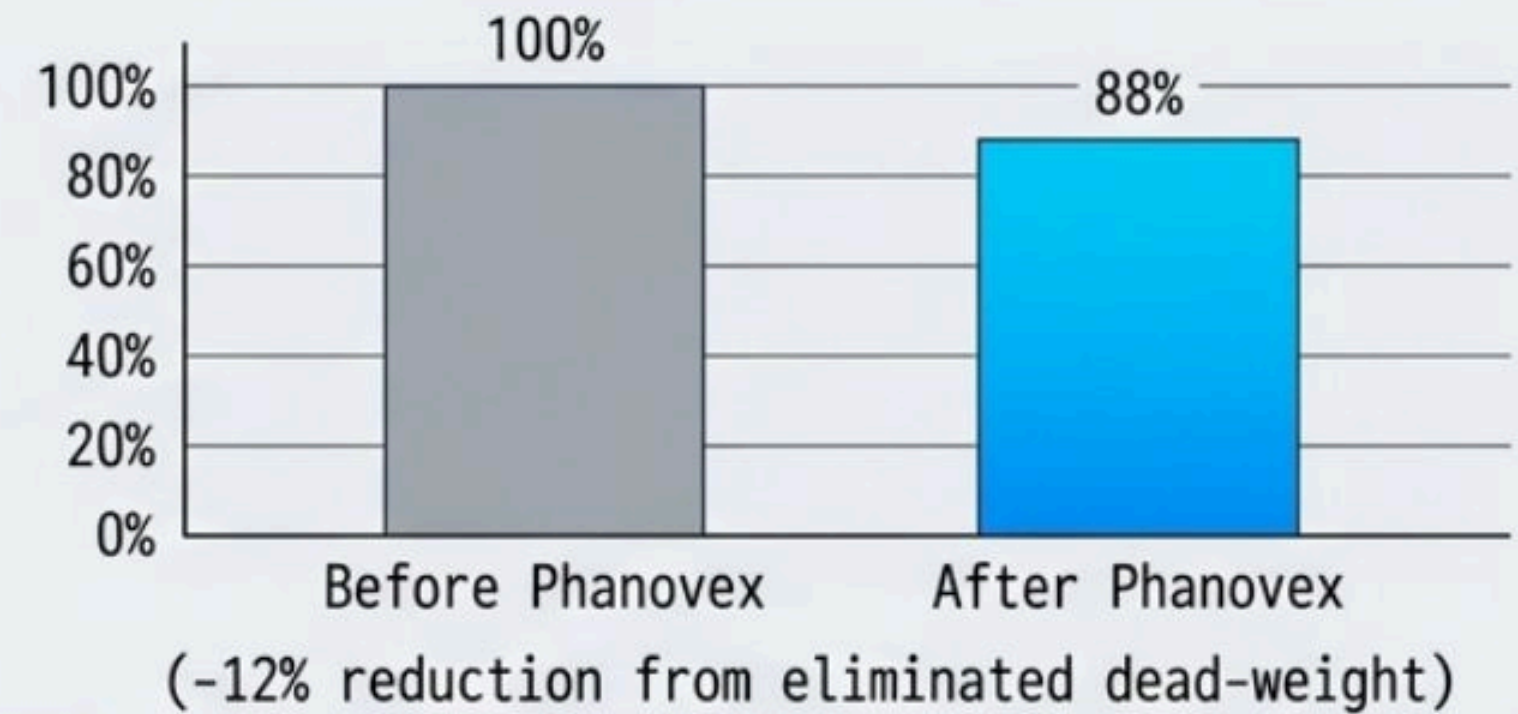


The Compounding Economic Return of Intelligence

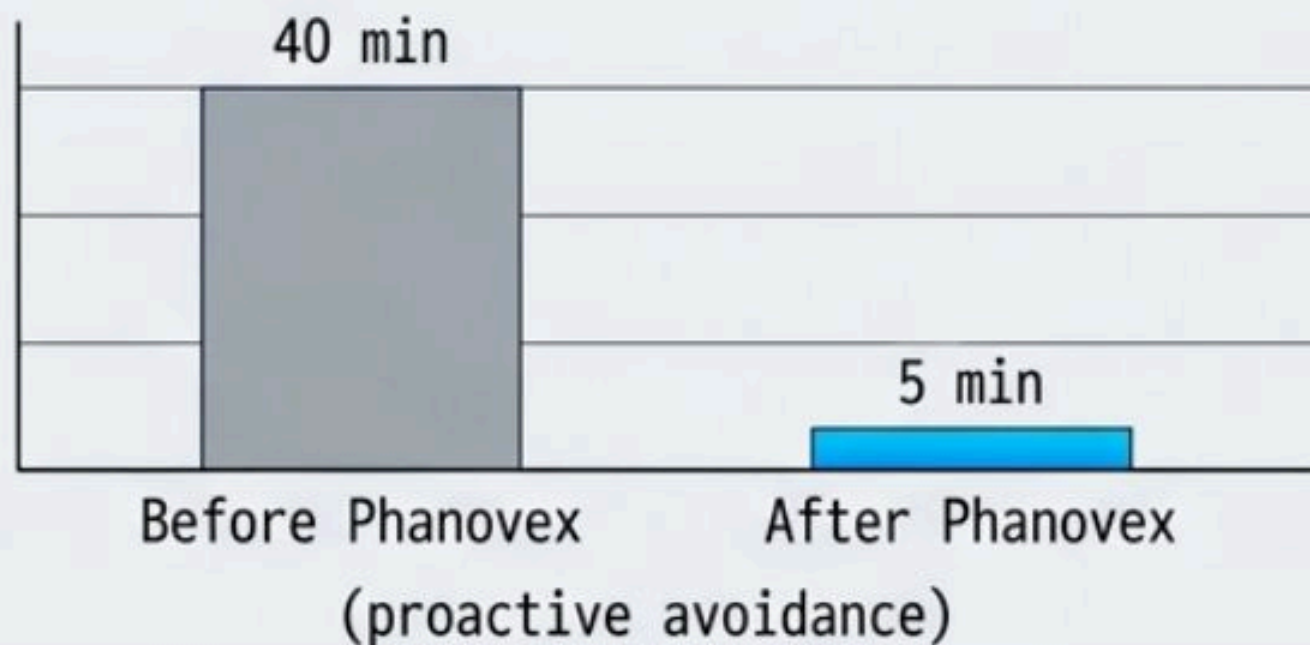
Payload Utilization



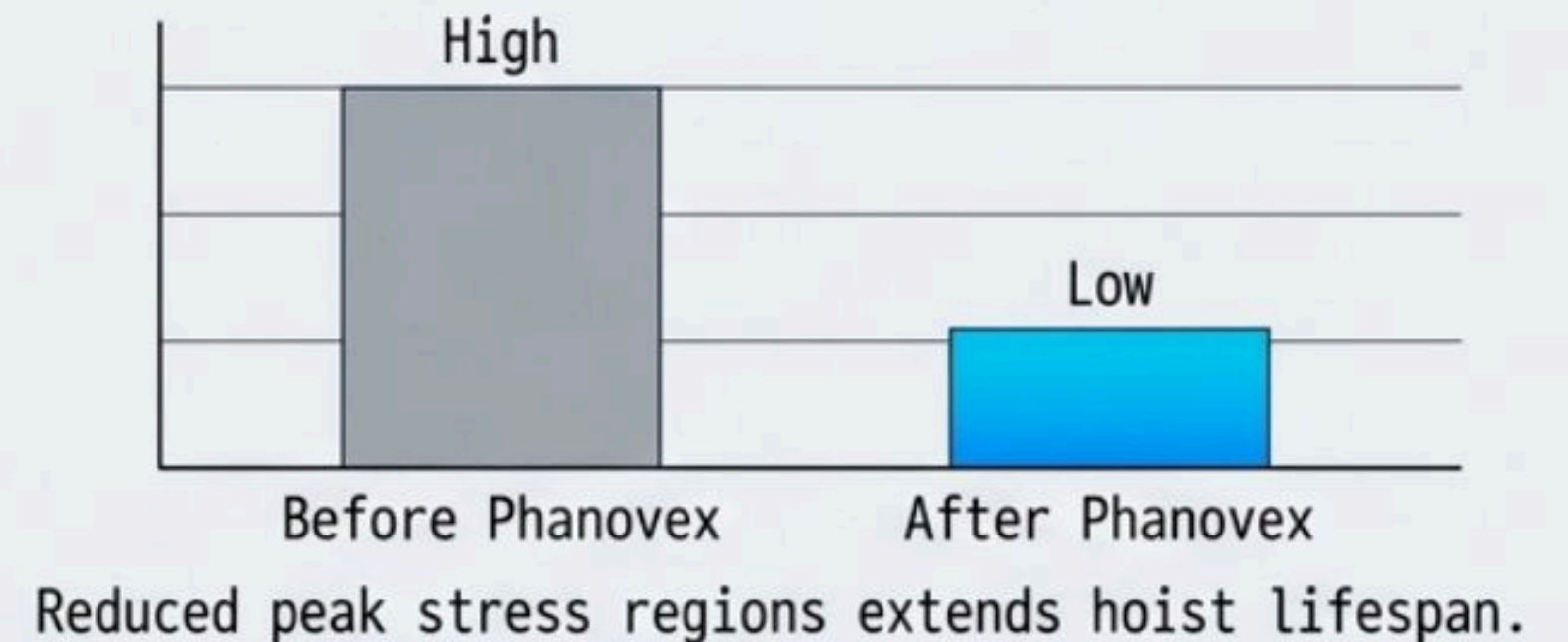
Fuel Consumption per Tonne



Cleaning Time per Shift



Equipment Health



Operational Impact & Value Proposition



Productivity

- Higher effective payload per cycle.
 - Faster load-haul-dump cycles.
 - Improved fleet utilization.
- 

Cost Reduction

- Lower fuel consumption from optimized routes.
- Reduced manual cleaning labor.
- Fewer rework trips.

Equipment Health

- Early detection of wear patterns.
- Predictive scheduling.
- Reduced structural stress on hoist/tray.

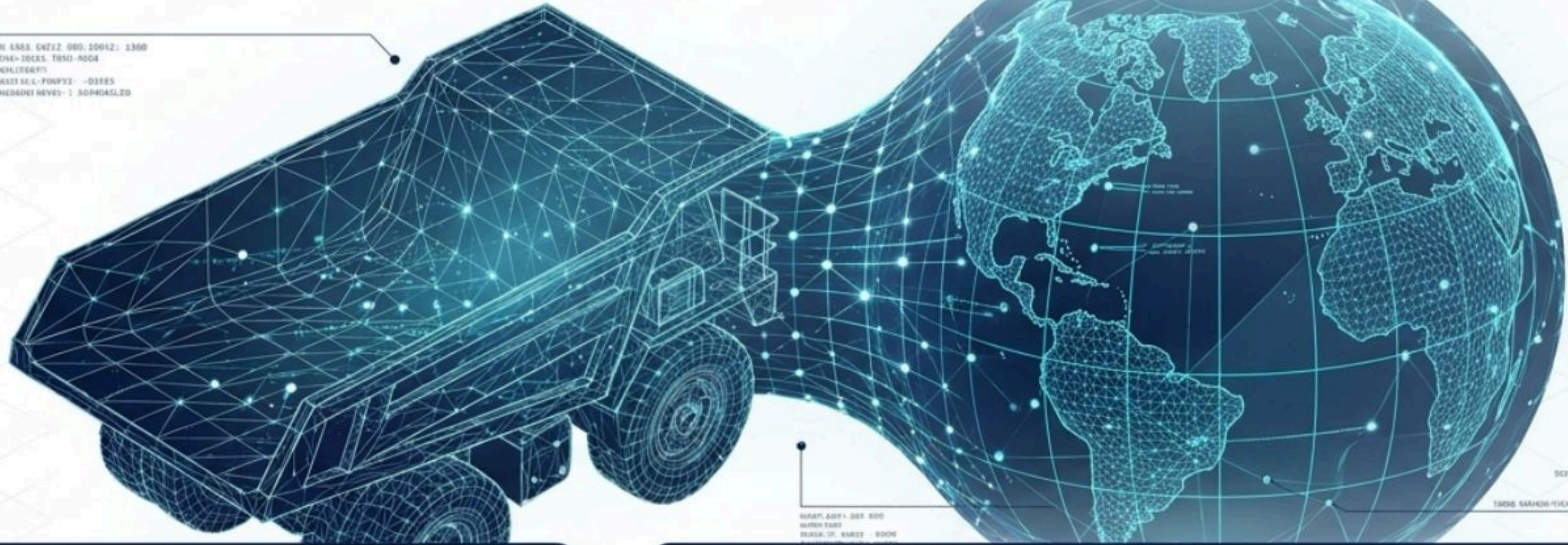
Deployment Advantages

- Retrofit compatible.
 - Scalable across both manual and CAT autonomous haul trucks.
- 

Every Haul Cycle Trains the Autonomous Future

“We solved it before the bed moves.”

DATA: 4555, 5X212, 080, 10012, 1300
NADNA-10000, 1000-1000
B: 1000/1000
S: 1000/1000, 1000/1000, 1000/1000
C: 1000/1000, 1000/1000, 1000/1000
...



Phanovex shifts carry-back from a reactive hardware nuisance to a predictable, solvable data problem.

By fusing Edge prediction with passive Biomimetic prevention and continuous Cloud learning, we ensure that every truck improves every other truck—paving the way for fully optimized, sustainable, and autonomous mining ecosystems.